

COMPUTER SCIENCE MSC CURRICULUM

Program: AUTONOMOUS SYSTEMS

ASSESSMENT TYPES:

E: Exam | PG: Practical Grade | XE: Lecture+Practice with Exam | XPG: Lecture+Practice with Practical Grade

=== COMPULSORY SUBJECTS (43 Credits) ===

--- SEMESTER 1 (21 Credits) ---

IPM-22AUTCVEG: 3D Computer Vision

2 Lecture, 0 Practice, 2 Laboratory, 2 Consultation | XE | 6 Credits

IPM-22AUTISPE: Image and Signal Processing (Lecture)

2 Lecture, 0 Practice, 0 Laboratory, 1 Consultation | E | 3 Credits | Prerequisite: IPM-22AUTISPG (weak)

IPM-22AUTISPG: Image and Signal Processing (Practice)

0 Lecture, 0 Practice, 2 Laboratory, 1 Consultation | PG | 3 Credits

IPM-22AUTIVSEG: Introduction to Vehicles and Sensors

2 Lecture, 1 Practice, 0 Laboratory, 1 Consultation | XE | 4 Credits

IPM-22AUTRMG: Research methodology

1 Lecture, 2 Practice, 0 Laboratory, 2 Consultation | XPG | 5 Credits

--- SEMESTER 2 (22 Credits) ---

IPM-22AUTAIPARG: Deep Reinforcement Learning

2 Lecture, 2 Practice, 0 Laboratory, 2 Consultation | XPG | 6 Credits

IPM-22AUTDAAE: Design and Analysis of Algorithms (Lecture)

2 Lecture, 0 Practice, 0 Laboratory, 2 Consultation | E | 4 Credits

IPM-22AUTIDSEG: Introduction to Data Science

2 Lecture, 0 Practice, 2 Laboratory, 2 Consultation | XE | 6 Credits

IPM-22AUTNMEG: Numerical Methods for Optimization

2 Lecture, 0 Practice, 2 Laboratory, 2 Consultation | PG | 6 Credits

IPM-22PIRG: Internship

0 Credits | Semesters: 2-4

=== COMPULSORY ELECTIVE SUBJECTS (Select 41 Credits) ===

--- SEMESTER 1 ELECTIVES ---

IPM-22AUTPHFTG: Human Factors in Traffic Environment (Practice)

0 Lecture, 2 Practice, 0 Laboratory, 0 Consultation | PG | 2 Credits | Semesters: 1,3

IPM-22AUTLFADE: Legal Framework for Autonomous Driving (Lecture)

2 Lecture, 0 Practice, 0 Laboratory, 0 Consultation | E | 2 Credits | Semesters: 1,3

IPM-22AUTPRFPG: Preparation course for master studies and developing learning skills

0 Lecture, 3 Practice, 0 Laboratory, 0 Consultation | PG | 2 Credits | Semester: 1

IPM-22AUTPME: Project Management (Lecture)

2 Lecture, 0 Practice, 0 Laboratory, 0 Consultation | E | 2 Credits | Semesters: 1,3

IPM-22AUTTAMEG: Topics in Applied Mathematics

2 Lecture, 2 Practice, 0 Laboratory, 1 Consultation | XPG | 5 Credits | Semester: 1

IPM-22AUTPAREG: Principles of artificial intelligence

2 Lecture, 0 Practice, 2 Laboratory, 2 Consultation | XE | 6 Credits | Semester: 1

--- SEMESTER 2 ELECTIVES ---

IPM-22AUTASTE: Advanced Software Technology (Lecture)

2 Lecture, 0 Practice, 0 Laboratory, 2 Consultation | E | 4 Credits | Semester: 2

IPM-22AUTAUSLAB1: AUS Lab I

0 Lecture, 3 Practice, 0 Laboratory, 2 Consultation | PG | 5 Credits | Semester: 2

IPM-23AUTG3DCVEG: Geometry of 3D Computer Vision

2 Lecture, 2 Practice, 0 Laboratory, 2 Consultation | XE | 6 Credits | Semester: 2

IPM-22AUTIVPFG: Image and Video Processing

2 Lecture, 2 Practice, 0 Laboratory, 2 Consultation | XE | 6 Credits | Semester: 2

IPM-24AUTIMREG: Intelligent Mobile Robots

2 Lecture, 2 Practice, 0 Laboratory, 2 Consultation | XPG | 6 Credits | Semester: 2

IPM-22AUTCGE: Computer graphics (Lecture)

2 Lecture, 0 Practice, 0 Laboratory, 1 Consultation | E | 3 Credits | Semester: 2 | Prerequisite: IPM-22AUTCGG (weak)

IPM-22AUTCGG: Computer graphics (Practice)

0 Lecture, 2 Practice, 0 Laboratory, 1 Consultation | PG | 3 Credits | Semester: 2

IPM-22AUTDAAG: Design and analysis of algorithms (Practice)

0 Lecture, 2 Practice, 0 Laboratory, 1 Consultation | PG | 3 Credits | Semester: 2

IPM-22AUTERTSEG: Embedded and Real-Time Systems

2 Lecture, 0 Practice, 2 Laboratory, 2 Consultation | XE | 6 Credits | Semester: 2

IPM-22AUTSQTE: Software quality and testing (Lecture)

2 Lecture, 0 Practice, 0 Laboratory, 1 Consultation | E | 3 Credits | Semester: 2 | Prerequisite: IPM-22AUTSQTG (weak)

IPM-22AUTSQTG: Software quality and testing (Practice)

0 Lecture, 0 Practice, 2 Laboratory, 1 Consultation | PG | 3 Credits | Semester: 2

--- SEMESTER 3 ELECTIVES ---

IPM-22AUTSCTE: System and Control Theory (Lecture)

2 Lecture, 0 Practice, 0 Laboratory, 1 Consultation | E | 3 Credits | Semester: 3 | Prerequisite: IPM-22AUTSCTG (weak)

IPM-22AUTSCTG: System and Control Theory (Practice)

0 Lecture, 0 Practice, 2 Laboratory, 1 Consultation | PG | 3 Credits | Semester: 3

IPM-23AUTSBCPARG: 3D point cloud processing and analysis

2 Lecture, 0 Practice, 2 Laboratory, 2 Consultation | XE | 6 Credits | Semester: 3

IPM-22AUTDNDEG: Deep Network Development

2 Lecture, 0 Practice, 2 Laboratory, 2 Consultation | XE | 6 Credits | Semester: 3

IPM-22AUTSSFEG: 3D Sensing and Sensor Fusion

2 Lecture, 2 Practice, 0 Laboratory, 2 Consultation | XE | 6 Credits | Semester: 3

IPM-24AUTNLEG: Machine learning

2 Lecture, 0 Practice, 2 Laboratory, 2 Consultation | XE | 6 Credits | Semester: 3

IPM-22AUTADLEG: Advanced Deep Network Development

2 Lecture, 2 Practice, 0 Laboratory, 2 Consultation | XE | 6 Credits | Semester: 3 | Prerequisite: IPM-22AUTDNDEG

IPM-22AUTAAE: Advanced Algorithms (Lecture)

2 Lecture, 0 Practice, 0 Laboratory, 1 Consultation | E | 3 Credits | Semester: 3 | Prerequisite: IPM-22AUTAAG (weak)

IPM-22AUTAAG: Advanced Algorithms (Practice)

0 Lecture, 0 Practice, 2 Laboratory, 1 Consultation | PG | 3 Credits | Semester: 3

IPM-22AUTAUSLAB: AUS Lab II

0 Lecture, 3 Practice, 0 Laboratory, 2 Consultation | PG | 5 Credits | Semester: 3

=== INNOVATION & ENTREPRENEURSHIP MODULE (Optional) ===

IPM-22fAEFBEG: I&E Basics

2 Lecture, 2 Practice, 0 Laboratory, 2 Consultation | XPG | 6 Credits | Semester: 1

IPM-22f&EIRDL1G: Business Development Lab I

0 Lecture, 2 Practice, 0 Laboratory, 2 Consultation | PG | 4 Credits | Semester: 1

IPM-22f&EIRDL2G: Business Development Lab II

0 Lecture, 2 Practice, 0 Laboratory, 2 Consultation | PG | 4 Credits | Semester: 2

IPM-22f&EIAOEEG: Innosocial aspects of entrepreneurship

2 Lecture, 2 Practice, 0 Laboratory, 2 Consultation | XPG | 6 Credits | Semester: 2

IPM-22f&ETS8G: Thematic Summer Schools with I&E project

1 Lecture, 1 Practice, 0 Laboratory, 2 Consultation | XPG | 4 Credits | Semester: 2

IPM-22f&ESTEG: I&E Study

2 Lecture, 2 Practice, 0 Laboratory, 2 Consultation | XPG | 6 Credits | Semester: 3

=== THESIS & ADDITIONAL ===

IPM-24AUTTHESIS: Thesis consultation

30 Credits | Semester: 4

IPM-22AUTERASMUS: Erasmus mobility

Maximum 24 Credits | Semester: 3

Elective Credits: 6 Credits | Semesters: 2,3

=== CREDIT SUMMARY ===

Compulsory Subjects: 43 Credits

Compulsory Electives: 41 Credits

Elective Subjects: 6 Credits

Thesis: 30 Credits

TOTAL DEGREE CREDITS: 120 Credits

Semester Credit Distribution:

Semester 1: 30 | Semester 2: 30 | Semester 3: 30 | Semester 4: 30

SPECIAL NOTES:

- ETF students must complete I&E module and all subjects marked with (*)
- ETF students complete internship and thesis work in their final semester